

FITBUDDY

AI-Powered Personalized Workout and Wellness Assistant

PHASE 06

Project Testing Phase

Define the verification approach and record the current automated test coverage and expected outcomes.

Prepared for phase-wise submission

30 September 2026

Phase objective: Define the verification approach and record the current automated test coverage and expected outcomes.

1. Testing Strategy

- Unit/functional verification focuses on health, home page, generation, feedback revision, and user listing.
- Test setup forces MOCK_AI=true and uses an in-memory SQLite database so tests do not require external AI access.
- FastAPI TestClient exercises the HTTP routes as an integrated application flow.
- The project also includes syntax compilation/packaging verification recorded in the README.

2. Test Cases

- TC-01 Health endpoint returns HTTP 200 and status=ok.
- TC-02 Home route returns HTTP 200 and renders the main call-to-action text.
- TC-03 Generation creates a user and returns a seven-day plan.
- TC-04 Feedback submission revises the plan and shows the revised message/content.
- TC-05 Users page renders stored users after plan generation.

3. Recorded Automated Result

- The packaged README records that the offline/mock automated test suite passed with 4 tests passing.
- The four test functions currently present are: test_health, test_home, test_generate_and_feedback, and test_users_page.
- This is a development verification record; live Gemini network calls were not executed in the packaging environment.

4. Expected Test Evidence

- Terminal output from pytest -q.
- Browser evidence of the generated plan.
- Browser evidence of the revised plan after feedback.
- Browser evidence of the users page.
- Swagger UI evidence from /docs.
- Optional screenshot of a healthy /health response.

5. Defect Prevention / Regression Checks

- Keep AI response schemas stable when prompts change.
- Run pytest after changes to routes, schemas, database models, or service interfaces.
- Retest environment setup after changing requirements or .env values.
- Check that exactly seven days are returned after any workout-generation prompt change.
- Verify that the mock flow remains functional even when Gemini is not configured.

Test Matrix

Test ID	Area	Expected result	Recorded status
TC-01	/health	200 + ok	Passed
TC-02	/	200 + home page	Passed
TC-03	Generation	Plan + user saved	Passed
TC-04	Feedback	Revised plan displayed	Passed

Test ID	Area	Expected result	Recorded status
TC-05	Users page	Stored user visible	Passed

Phase Deliverables / Evidence

- Test strategy and matrix
- Automated verification record
- Regression and defect-prevention checklist

Submission Note

This phase document is intended to be submitted together with the corresponding FitBuddy source project and evidence captured during execution. Keep API keys, personal credentials, and other secrets out of screenshots and submissions.